

Vons: A Compact, Host-Controlled Decision Component for Agent Workflows

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Preprint v1.1, September 2026

Abstract

Agents built around frontier language models repeatedly make small, bounded decisions: which of a few candidate actions to take, whether a precondition holds, or whether to ask before acting. We ask whether a compact local model can serve such a decision inside a host-controlled workflow, and what breaks under the constraints we tested. Vons is a decision component consisting of a bounded decision contract, a shared frozen encoder (a four-layer BERT) with two decision heads—a one-pass *Direct* head and a DDIM *Diffusion* head—and a host-side adapter in which the model only proposes and the host alone validates, authorizes and executes. Two bodies of evidence are kept apart. The frozen v1 packet (2026-09-24) is mostly negative: a single-label synthetic split on which Direct’s answerable accuracy of 1.0 equals that of a constant predictor while the four-step Diffusion head reached 0.011–0.060; a temperature fit that stopped at its search boundary; and, on the original Mind2Web test splits, 37,512 exact- k selection attempts with no accepted selection because 84.7% exceeded the 512-token input budget. Its positive findings are narrow: a 46,001,618-byte shared bundle under a 64 MiB design target, exact CPU graph-partition parity in 80 of 80 comparisons, and a CPU shape decomposition in which padding, not the choice of head, dominates request time. A post-freeze remediation round (evidence dated 2026-09-25, supplied through a Codex-verified handoff and not independently rerun by us) addressed each failure: dynamic per-request padding with a re-verified partition; a balanced four-class synthetic set with vector-scaling calibration whose temperatures are interior (calibrated test ECE 0.0366, macro-F1 1.0 against a constant-majority 0.1017); a token-conditioned Diffusion head that reaches answerable accuracy 1.0 on that synthetic test; and label-free Mind2Web context compression that removed every overflow (0 of 37,512) and produced 323 correct selections among 5,553 recalled rows (conditional selection accuracy 0.0582) under an uncalibrated gate. These remain synthetic, local-CPU or conditional-selection results. We claim no external generalization, no superiority of either head, no calibrated deployment and no production readiness.

Licence and status. The article text and its original figures are licensed under CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>). The licence does not cover the Vons software, quoted source excerpts, third-party assets, raw benchmark data or private verification records (Section 11). This is a research preprint (v1.1, September 2026); it has not been peer reviewed. arXiv submission 8127259 is on hold, and this version does not itself update or publish it. Frozen v1 measurements were read from one hash-verified evidence packet (2026-09-24); post-freeze v2 values (2026-09-25) come from a Codex-verified evidence handoff whose files are not part of that packet and were not accessed or re-run by the drafting assistant (Section 5.1, Section 7). Corresponding author: osworld@inlevel9.com.

1 Introduction

A frontier-model agent spends much of its budget on decisions whose answer space is small and known in advance: pick one of up to 32 candidate elements, confirm that a precondition holds, or decide to ask for clarification. Serving such a step with a compact local model is attractive for cost, latency and

privacy, and it is a natural complement to routing and cascading schemes that send only hard queries to a large model [1], [2]. It also raises three separable questions.

1. **Capability.** Does the compact model choose as a reference would on the decisions that actually occur?
2. **Cost.** How large are the model assets, and how long does a request take on the target runtime? For Vons that runtime is a browser running ONNX Runtime Web [3] with a WASM or WebGPU execution provider.
3. **Authority.** Can the model’s output ever cause an action that the host did not authorize?

Vons answers the third question by construction: the model only proposes, and the host derives every consequential action (Section 3). The first two questions are empirical. This paper reports what the frozen v1 evidence packet supports on them, gives the negative results the same weight as the positive ones, and then reports—separately labelled—the post-freeze v2 remediation evidence that addressed each recorded failure (Section 7).

Contributions.

1. A bounded decision contract and a host-side adapter (KASI) in which confirmation is derived from host-owned risk metadata and exact-call consent, never from the model (Section 3).
2. A shared-encoder model with two decision heads, packaged as a single model-only bundle under a 64 MiB design target, with exact CPU graph-partition parity (Section 4 and Section 6.4).
3. A CPU shape decomposition showing that input padding, not the choice of head, dominates request time at the tested shapes (Section 6.5).
4. Informative negative results: a synthetic split that cannot distinguish learning from a constant predictor, a calibration fit that stopped at its boundary, and a Mind2Web diagnostic in which no selection was accepted because most inputs exceeded the token budget (Section 6.1 to Section 6.3).
5. An evidence discipline in which every reported number is bound to a hash-verified source field, with measured, historical, derived, design-target, planned, unavailable and post-freeze quantities kept apart (Section 5.1 and Appendix B).
6. A post-freeze remediation round on the four recorded failures—dynamic padding with re-verified parity, a balanced synthetic set with vector-scaling calibration, a token-conditioned Diffusion head, and Mind2Web context compression—reported against the frozen baseline in one table (Section 7).

What we do not claim. We do not claim that Vons generalizes to external tasks, that either head is better, that the calibration holds beyond the synthetic split, that the browser latencies reflect the current software, that Mind2Web conditional selection implies browser task success, or that the system is ready for production use. Post-freeze values were not independently rerun by us.

2 Related Work

Tool-using and web agents. ReAct interleaves reasoning traces with actions in language models [4], and Toolformer teaches a model to emit API calls inline [5]. Mind2Web provides tasks on real websites with crowdsourced action sequences for generalist web agents [6]; SeeAct grounds a multimodal model’s plans into element-level actions on the same benchmark [7], and WebArena supplies self-hosted environments for end-to-end evaluation [8]. Our external diagnostic uses the original Mind2Web test release. Vons addresses a narrower sub-problem than these systems: a single bounded choice inside a host-controlled loop, where the surrounding planner is out of scope.

Authority and risk in tool use. ToolEmu emulates tool execution to surface the risks that agents incur when they act with real consequences [9]. Vons takes the complementary, structural position that the model should never hold execution authority: the host owns the tool registry, the risk metadata

and the consent binding, and the model’s output is advisory (Section 3). We present this as a design property of the frozen source, not as a measured safety outcome.

Compact encoders and distillation. Knowledge distillation transfers a large model’s behaviour into a smaller one [10]; DistilBERT and TinyBERT apply it to BERT-style encoders [11], [12], and pre-training remains important for compact models [13]. Vons uses one such compact pre-trained encoder [13], [14] without task-specific distillation. No local model was admitted as a distillation teacher in this work (Section 6.7).

Diffusion decoding. Denoising diffusion models [15] and the non-Markovian DDIM sampler that permits few-step generation [16] have been applied to discrete outputs through continuous relaxations [17]. The Vons Diffusion head applies four DDIM steps to candidate scores; its configuration is a first attempt with known design limitations (Section 4), and its results should not be read as a verdict on diffusion decoding.

Calibration and abstention. Temperature scaling is a single-parameter post-hoc calibration method [18]; expected calibration error (ECE) and the Brier score are standard summaries [19], [20]. Selective classification trades coverage for risk through a reject option [21], [22], and the maximum softmax probability is a common abstention signal [23]; language models’ own confidence has also been studied [24]. Vons reports coverage and selective risk alongside accuracy for this reason.

Vendor workflow evaluations. TypeSafe publishes a four-category workflow evaluation whose reference labels average GPT-6 Astra and Claude Fable 5.1 at high thinking [25], [26]. Scores there measure agreement with model-derived references rather than independently labelled truth. No comparison with that evaluation has been run for Vons, and no vendor figure is reproduced here.

3 The Decision Contract and the Host Boundary

Vons is a decision component, not an agent. A host application (for example, an agent runtime that uses a frontier model for planning) reaches a point where it must choose among a small, explicit set of options. It sends Vons a structured request. Vons returns a probability over the options, an answerability estimate, and either a choice or an abstention. The host then decides what, if anything, to do.

General decision contract. A request carries a structured state and 1–8 typed questions, each with 2–32 options, under a 512-token input budget. The response carries a choice with a probability vector and an answerability estimate, or an abstention. Score-type questions are not supported by the current candidate-selection ONNX runtime; an expected-ordinal score output is planned for an evaluation adapter and is not part of the implemented response. The Python validator estimates the token budget from characters, whereas the TypeScript SDK uses a tokenizer budget, so the two implementations do not yet form one harmonized public contract.

Host boundary (KASI adapter). For tool-using workflows, a separate host-side adapter called KASI maps the general contract onto five public actions: *call*, *clarify*, *confirm*, *refuse* and *respond*. KASI is kept separate from the general contract. Table 1 summarizes who decides what. The default confidence threshold is 0.55, and the Python adapter delegates the derivation to a single contract function. These are properties of source code and documentation; no adapter audit log from a live workflow exists, so boundary behaviour in deployment is UNAVAILABLE.

Table 1. Division of authority in the KASI host adapter. Implemented in the frozen source; not measured on live workflows.

Decision	Made by	Rule in the frozen source
Propose a call, clarification, refusal or response	Model	Proposal only; never executed by the model
Whether a tool may be used	Host	Unregistered tools are refused
Risk of a tool	Host registry	Missing or unknown risk fails closed
Whether to act at all	Host	Confidence below the threshold yields <i>clarify</i>
Whether confirmation is needed	Host	Required unless both the host-registered and the supplied advisory risk rank below <i>high</i>
Consent	User, via host	Bound to the complete call, including arguments
Execution	Host	The host is the only execution boundary

4 Model and Implementation

Encoder and heads. Both heads share the frozen encoder `google/bert_uncased_L-4_H-256_A-4` [13], [14] at a pinned revision (Appendix B), with a 512-token maximum and 32 option slots. The *Direct* head scores all candidates in one pass. The *Diffusion* head denoises candidate scores with four DDIM steps [16]. The historical Diffusion head has three known design limitations: it conditions on the pooled representation only, it uses a weak 32-step terminal noise schedule, and it applies a softmax over denoised one-hot scores. Its results therefore do not establish that diffusion decoding is inferior in general. A token-conditioned redesign with eight inference steps was trained after the freeze (Section 7.3).

Software. The implementation consists of a Python package (contract, data generation, training, evaluation, calibration, bundle manifest, KASI adapter and Mind2Web tooling), a TypeScript SDK (contract validation, KASI host policy and a browser runtime on ONNX Runtime Web; the historical browser runs recorded version 1.30.0), and exported ONNX graphs.

Shared bundle. The *shared-v1* bundle partitions the two existing full graphs into one encoder graph and two head graphs, with no training or parameter change (Section 6.4). Its recorded release status is “research candidate; parity and notices required before release”.

5 Evaluation Protocol

5.1 Evidence basis and labelling

All measurements come from one frozen evidence packet (schema `vons.publication-evidence/v2`, frozen 2026-09-24). Before drafting, we verified the packet archive and manifest digests, all 131 listed files by SHA-256 and byte count, and all 258 claim rows supplied with the packet. We also recomputed the six synthetic-pilot metric summaries from the saved predictions and test labels; the largest absolute difference from the recorded values was 2.2×10^{-16} . The packet records `git_revision = null` because the workspace had no Git repository at the 2026-09-24 freeze; provenance therefore rests on content hashes (Appendix B). A repository has since been designated (Section 11), but no later commit is bound to any measurement reported here.

We did *not* re-run training, model inference, browser tests or the Mind2Web evaluation. The restricted Mind2Web test data and the model weights are deliberately absent from the packet; their manifests, hashes and aggregate reports are present. Our Mind2Web statements therefore rest on verification of the aggregate reports, not on independent re-evaluation.

Post-freeze evidence. A second body of evidence, dated 2026-09-25 and indexed by `reports/goal-v2/evidence-summary.json` (SHA-256 in Table 12), was produced after the freeze and reached us as a Codex-verified handoff of file paths, digests and values. It is not part of the frozen packet; we did not access,

recompute or re-run any of it. Every quantity from it carries the label `POST-FREEZE (HANDOFF)` and is confined to Section 7, the comparison table there, and the sections that interpret it.

Throughout, quantities carry one of the following labels. `MEASURED`: read from a named field of a hash-verified report. `MEASURED (HISTORICAL)`: measured on an earlier configuration that is not the current software. `DERIVED`: arithmetic on measured fields, recomputed for this paper. `DESIGN TARGET`: an intended property, never an outcome. `PLANNED`: intended future work, not implemented or not run. `UNAVAILABLE`: not measured, or recorded as null in the source. `AUTHOR DECISION`: a licensing or status choice made after the evidence was frozen. `POST-FREEZE (HANDOFF)`: a value measured after the freeze and supplied through the Codex-verified handoff, not independently rerun. Values in tables are rounded to the precision shown; the full-precision values are in the named packet fields.

5.2 Synthetic pilot data

The synthetic test split has 200 rows. A separate calibration split, filtered from the same generated source, also has 200 rows. Both splits have the same composition: 182 answerable rows, *all labelled* `call_tool`, 18 unanswerable rows, and four options per row. A constant `call_tool` predictor that never abstains would therefore reach answerable accuracy 1.0 on either split. Synthetic accuracy is not evidence of learned decision quality; we treat it only as a check that the pipeline runs end to end.

5.3 Metrics

Metrics follow the packet’s `metrics/v2` definitions.

- **Answerable accuracy**: correct covered predictions divided by all answerable rows; missing, error and abstained predictions count as misses.
- **Coverage**: covered predictions divided by all rows.
- **Selective risk**: errors divided by covered predictions; a covered prediction on an unanswerable row is an error.
- **ECE**: ten equal-width bins over covered predictions, computed on the recorded confidence. The recorded confidence is the maximum choice probability multiplied by the sigmoid of the answerability output, as computed in the frozen prediction loop; it differs from the choice probabilities used by Brier and NLL.
- **Brier** and **NLL** are means over a different population from ECE: the answerable rows whose prediction has no error and carries a valid probability vector. Abstained predictions are included when a valid vector exists. Per row, Brier is the multiclass score summed over the row’s options, $\sum_o (p_o - \mathbf{1}[o = \text{label}])^2$, so its range is 0–2; NLL is $-\log \max(p(\text{label}), 10^{-12})$. An empty population yields null (frozen `vons/evaluation.py`, metric definitions and `_nll/_brier`).

5.4 Mind2Web evaluation-only protocol

Data. We used the original Mind2Web test release [6] at a pinned dataset revision and repository revision (Appendix B). The licence is recorded as CC-BY-4.0 with research-only, no-training and evaluation-only restrictions. The train split was not acquired, so no train/test overlap audit exists. In total, 1,341 tasks and 9,378 action rows were normalized.

Retrieval and prediction. Candidates are ranked by lexical BM25 [27] ($k_1 = 1.2$, $b = 0.75$) over the released candidate pool, using the task goal and the teacher-forced official history strictly before the current action. The shared-v1 Direct model ran exact- k action-target selection for $k \in \{5, 10, 20, 32\}$, with a strict 512-token budget, no truncation, no calibration, and the historical uncalibrated gates of 0.55 confidence and 0.5 answerability.

Replay and intervals. A replay recomputes outcome counts from saved predictions. Task-macro intervals use 1,000 task-level bootstrap draws with seed 7. We report candidate recall and conditional selection accuracy separately. Browser task success and end-to-end DOM retrieval are not evaluated.

5.5 Timing protocols

CPU shape decomposition. This ran on CPU with ONNX Runtime 1.30.0 on macOS 26.6.2 (arm64), using the historical full graphs rather than shared-v1. The design used four synthetic cases, 20 repeats and 3 session starts, with 5 warm-up requests excluded per session, giving 240 samples per cell. The same live texts and masks were used across paired cells. Four cells were measured: (A) Direct at 4 slots $\times$ 64 tokens, (B) Direct at 32×64 , (C) Direct at 32×512 and (D) Diffusion at 32×512 .

Browser (historical). The historical Direct full graph was timed with 4 candidates in 4 padded slots at 512 tokens, with ten warm-up requests excluded per page/session start. The execution provider was *requested*, not observed: the observed provider is null and the requested provider is not proven. Fresh page sessions are not independent browser-process launches, and the page was not cross-origin isolated.

5.6 Local-model screen

Three local models (Qwen, Gemma and Muse Glimmer; Table 8) were each given the same 200 synthetic test rows under role metadata “writer”, “judge” and “analyst”. A later read-only audit bound all 600 responses to prompt and response hashes and made no new model calls. The three roles received identical prompt sets.

6 Results

6.1 Synthetic pilot

Table 2 reports the two heads separately; we do not rank them.

Direct. At every seed the Direct head covered all 200 rows and answered every answerable row correctly. Its selective risk of 0.09 is exactly the 18 covered unanswerable rows. Because of the single-label split, this shows only that the pipeline runs end to end.

Diffusion. The Diffusion head covered 22–58 rows per seed and answered 2–11 answerable rows correctly. Its answerable accuracy of 0.0110–0.0604 is far below what a constant predictor attains on this split. This is a negative result for this configuration, not for diffusion decoding in general.

Head timing. The pilot reports also record decision-head latency with precomputed encoder embeddings. That quantity is neither request latency nor browser latency, and we do not report it as either.

6.2 Calibration (Direct, seed 7)

In the frozen source, calibration first fits a temperature and then fits an abstention threshold on temperature-scaled confidence for a design-target selective risk of 0.05, both on the separate calibration split (Table 3).

Table 2. Synthetic pilot, 200-row test split, metrics/v2 (MEASURED). The two heads are reported separately and are not compared: the split is single-label (Section 5.2), so answerable accuracy is uninformative about learning.

Head	Seed	Ans. acc.	Macro-F1	Coverage	Sel. risk	NLL	Brier	ECE
Direct	7	1.0000	1.0000	1.000	0.0900	0.1053	0.0139	0.2163
Direct	17	1.0000	1.0000	1.000	0.0900	0.0871	0.0097	0.1154
Direct	27	1.0000	1.0000	1.000	0.0900	0.1013	0.0129	0.2881
Diffusion (4 DDIM steps)	7	0.0275	0.0134	0.155	0.8387	1.8031	0.9318	0.4483
Diffusion (4 DDIM steps)	17	0.0604	0.0285	0.290	0.8103	1.8147	0.9370	0.4602
Diffusion (4 DDIM steps)	27	0.0110	0.0054	0.110	0.9091	1.8165	0.9368	0.5088

Table 3. Calibration fit and calibrated test result for the Direct head, seed 7. Values rounded to four decimals.

Quantity	Value	Label
Temperature-fit rows / threshold-fit rows	182 / 200	MEASURED
Fitted temperature (search range 0.05–20)	0.05 (at lower bound)	MEASURED
Abstention threshold	0.7558	MEASURED
Target selective risk	0.05	DESIGN TARGET
Calibration-split risk at threshold	0.0	MEASURED
NLL before / after (calibration split)	0.1064 / 0.0	MEASURED
Test coverage / selective risk / answerable accuracy	0.91 / 0.0 / 1.0	MEASURED
Test abstentions / covered unanswerable rows	18 / 0	MEASURED
Test ECE	0.2233	MEASURED

With the temperature at its lower bound the choice probabilities saturate: test Brier is 2.29×10^{-51} , test NLL is 0.0, and $1 - \overline{\max p}$ over covered rows is 0.0. Every covered test prediction is correct, so the calibrated ECE equals $1 - \overline{\text{confidence}}$ exactly, and that value is set by the answerability factor, not by the temperature. The 0.91 coverage corresponds exactly to abstaining on the 18 unanswerable rows.

This is an empirical result on one synthetic split for one seed of one head. It is not a risk guarantee and not calibration on any external task. Diffusion calibration is UNAVAILABLE, immutable checkpoint identity is absent from the prediction configurations, and calibrated browser parity is open.

6.3 Mind2Web: candidate recall and selection

Figure 1 and Table 4 show every split/ k cell.

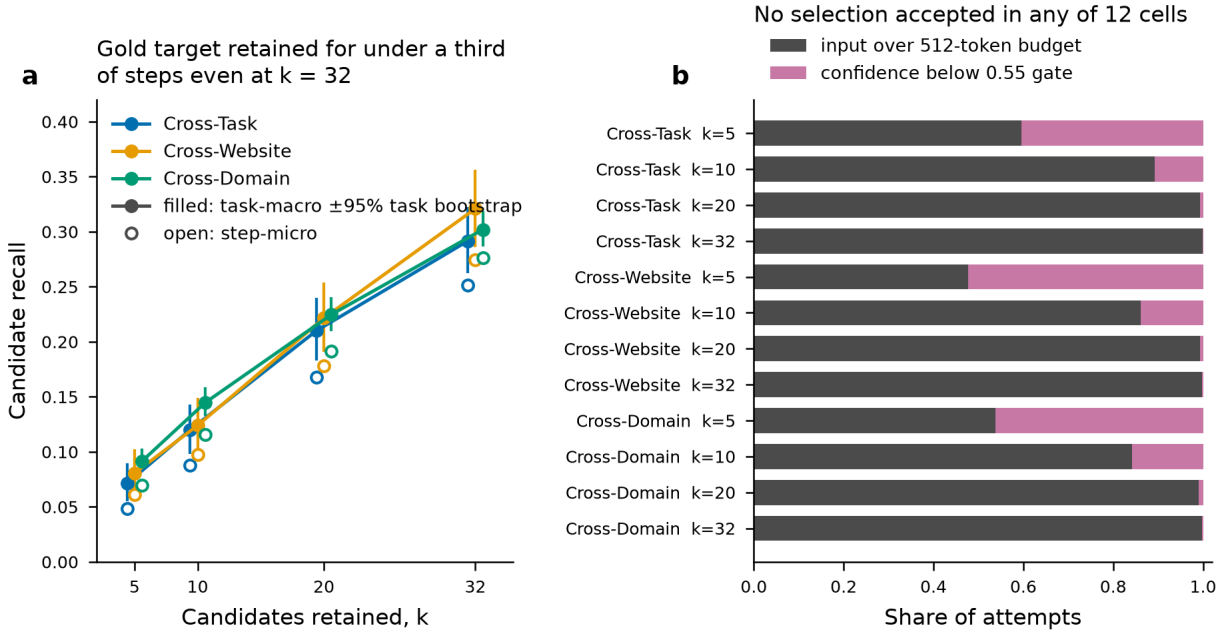


Figure 1. Mind2Web diagnostic on the original test splits (strict 512-token budget; uncalibrated 0.55/0.5 gates). (a) Candidate recall of the lexical BM25 retriever over the released pool; filled markers are task-macro recall with 95% task-bootstrap intervals, open markers are step-micro recall, and points are offset horizontally for legibility. (b) Outcome of every Direct selection attempt; accepted selections are zero in every cell.

Table 4. All twelve Mind2Web split/ k cells (MEASURED). “Overflow” means the input exceeded the 512-token budget; “abstain” means confidence fell below the 0.55 gate. Conditional selection accuracy is 0.0 in every cell. Rows per split (with a positive candidate in the pool): Cross-Task 2,094 (1,975); Cross-Website 1,373 (1,314); Cross-Domain 5,911 (5,590).

Split	k	Step-micro recall	Task-macro recall	95% CI (task-macro)	Recalled	Overflow	Abstain	Accepted
Cross-Task	5	0.0486	0.0713	[0.0549, 0.0894]	96	1,246	848	0
Cross-Task	10	0.0876	0.1199	[0.0983, 0.1429]	173	1,867	227	0
Cross-Task	20	0.1676	0.2100	[0.1833, 0.2397]	331	2,081	13	0
Cross-Task	32	0.2516	0.2916	[0.2623, 0.3223]	497	2,092	2	0
Cross-Website	5	0.0609	0.0806	[0.0615, 0.1025]	80	655	718	0
Cross-Website	10	0.0974	0.1240	[0.0992, 0.1489]	128	1,182	191	0
Cross-Website	20	0.1781	0.2216	[0.1908, 0.2537]	234	1,364	9	0
Cross-Website	32	0.2747	0.3210	[0.2860, 0.3561]	361	1,370	3	0
Cross-Domain	5	0.0698	0.0916	[0.0819, 0.1027]	390	3,180	2,731	0
Cross-Domain	10	0.1159	0.1447	[0.1328, 0.1587]	648	4,980	931	0
Cross-Domain	20	0.1916	0.2248	[0.2099, 0.2405]	1,071	5,852	59	0
Cross-Domain	32	0.2762	0.3021	[0.2869, 0.3188]	1,544	5,900	11	0

Totals. Across the twelve cells there were 37,512 selection attempts. Of these, 31,769 (84.7%) were input overflows and 5,743 were confidence abstentions; no record was missing and no selection was accepted.

Consequences. Conditional selection accuracy is 0.0 in every cell under the frozen metric, which keeps every recalled failure in the denominator, and acceptance coverage is 0.0. Complete-case conditional accuracy is undefined, because no selection was accepted. Each k has 499 rows without a positive candidate, counted separately.

Recall. Candidate recall rises with k . At $k = 32$, step-micro recall is 0.2516–0.2762 and task-macro recall is 0.2916–0.3210 across splits. This is a property of the lexical retriever and the released pool, not of the Vons model, and because history is teacher-forced it is not end-to-end retrieval.

Result. Under this protocol the shared-v1 Direct model is not useful on Mind2Web. The failure is dominated by the 512-token input budget: most attempts never reach inference.

6.4 Model assets and CPU graph parity

Table 5 lists the shared-v1 model-only bundle.

Table 5. Shared-v1 model-only bundle (MEASURED runtime and notices excluded).

Asset	Role	Bytes
encoder.onnx	encoder graph	553,978
encoder.onnx.data	encoder weights	44,423,168
direct.onnx	Direct graph	22,843
direct.onnx.data	Direct weights	132,096
diffusion.onnx	Diffusion graph	82,287
diffusion.onnx.data	Diffusion weights	74,752
tokenizer/tokenizer.json	tokenizer	711,396
tokenizer/tokenizer_config.json	tokenizer	379
calibration.json	calibration	233
config.json	configuration	486
Total model assets		46,001,618
Model-asset limit (64 MiB)	DESIGN TARGET	67,108,864

The shared bundle totals 46,001,618 bytes (43.87 MiB), below the 67,108,864-byte model-asset design target. Runtime bytes are null because no runtime is included. The two original full bundles are 45,853,779 bytes (Direct) and 45,899,284 bytes (Diffusion). Each is below the target on its own, but shipped together they total 91,753,063 bytes, which is 24,644,199 bytes over it (DERIVED).

Parity. Graph-partition parity compared the shared-v1 graphs with the full graphs on CPU (CPUExecutionProvider, ONNX Runtime 1.30.0) over 20 cases and 80 comparisons at $\text{atol} = \text{rtol} = 10^{-5}$. All 80 passed, with a maximum absolute difference of 0.0 for scores and answerability. The cases covered 2/4/8/16/32 candidates, batch sizes 1 and 2, and Diffusion noise seeds 7/17/27. This is graph-partition equivalence only; it says nothing about the tokenizer, the browser, calibration or latency.

6.5 CPU shape decomposition

Figure 2 and Table 6 give the four matched CPU cells.

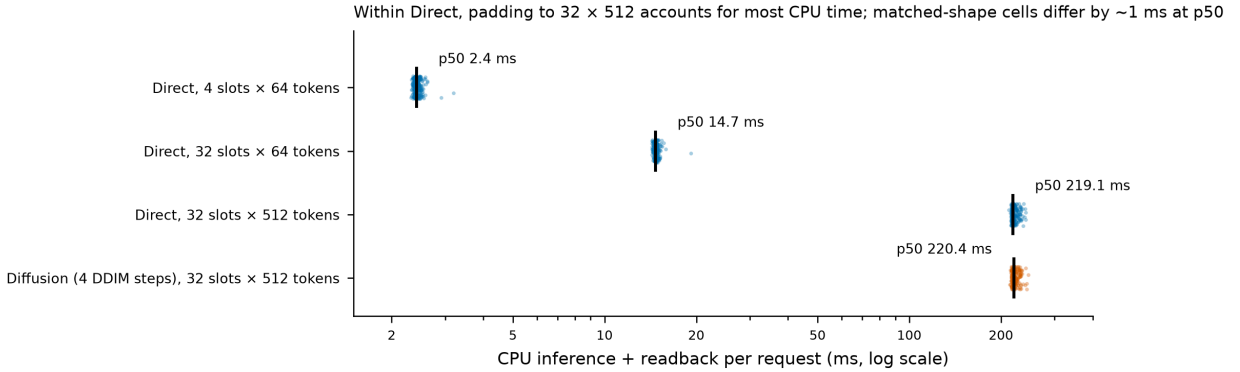


Figure 2. Per-request CPU inference plus readback time for four matched cells (240 samples each; four synthetic cases, three session starts, 20 repeats, five warm-ups excluded). Black ticks mark the median. Historical full graphs on ONNX Runtime 1.30.0 CPU; not a browser measurement.

Table 6. CPU shape cells, inference plus readback (MEASURED).

Cell	Configuration	Samples	p50 (ms)	p95 (ms)
A	Direct, 4 slots $\times$ 64 tokens	240	2.416	2.535
B	Direct, 32 slots $\times$ 64 tokens	240	14.705	15.154
C	Direct, 32 slots $\times$ 512 tokens	240	219.078	232.687
D	Diffusion (4 DDIM steps), 32 slots $\times$ 512 tokens	240	220.367	232.473

Within Direct, expanding 4 slots to 32 at 64 tokens adds 12.289 ms at p50 (B – A), and expanding 64 tokens to 512 at 32 slots adds 204.373 ms (C – B). The matched-shape difference D – C is 1.289 ms

at p50 (all DERIVED). The latter is a system difference that still includes architectural differences; it is neither the cost of DDIM alone nor a method-only speed comparison. The decomposition covers four cases on one CPU, and matched-padding cells in the browser remain open.

6.6 Historical browser timings (Direct full graph)

Table 7 reports the only browser measurements in the packet. They are historical and use the Direct full graph, not the current SDK or the shared-v1 bundle.

Table 7. Historical browser measurements of the Direct full graph (MEASURED, HISTORICAL). Both columns use 4 candidates $\times$ 512 tokens, 500 timed samples, five page/session starts and ten warm-ups excluded per start. “Requested” means the execution provider was requested by the page, not observed.

Quantity	Requested WASM	Requested WebGPU
Successful / failed samples	500 / 0	500 / 0
Page/session starts	5	5
Inference + readback p50 / p95 (ms)	2207.0 / 2470.2	91.0 / 130.5
Total request p50 / p95 (ms)	2207.4 / 2471.0	91.4 / 132.0
Tokenize p50 (ms)	0.3	0.2
JS heap used after (bytes; JS-heap scope only)	11,811,939	57,884,604
Requested provider proven	false	false
Process-level cold start claimable	false	false

These are request configurations measured on one Mac. The executing provider and graph partition are unverified, so the difference between the columns cannot be attributed to WebGPU execution and no speed ratio is claimed. Process and GPU peak memory are UNAVAILABLE the JS-heap figures describe only that heap. The current SDK and shared-v1 bundle were not timed, the Diffusion head has no browser cells, and the one attempt at a browser numerical-parity matrix stalled without a raw report and is not a pass.

6.7 Local-model screen

Table 8. Historical local-model screen (MEASURED, HISTORICAL 200 synthetic rows each; no new model calls). ^a Whether the local model conversion carries a licence field; not a statement about upstream licence terms.

Model (local tag)	Role	Params, quant.	Rows	Strict JSON	Null on abstain	Known-label acc.	Licence field ^a
qwen3.8:27b-mlx	writer	27.8B, nvfp4	200	200	200	0.00	true
gemma4:31b	judge	31.3B, Q4_K_M	200	200	200	0.00	true
muse-glimmer:30b-mlx	analyst	32.3B, nvfp4	200	200	182	0.09	false

On the 182 answerable rows labelled `call_tool`, all three models chose `respond_directly`. The saved prompts neither identify a required tool nor state a rule that would make `call_tool` decidable, so these low accuracies do not rank the models. Muse Glimmer’s 0.09 comes from its 18 abstention flags; on those rows it kept a non-null `clarify` choice, so only 182 of its 200 rows satisfy the null-on-abstention contract. No output was used for training, no local model was admitted as a teacher, and the upstream revisions of all three local conversions are null.

7 Post-freeze Remediation: v2 Evidence (2026-09-25)

Everything in this section carries the label POST-FREEZE (HANDOFF): it was measured after the evidence freeze, is indexed by `reports/goal-v2/evidence-summary.json`, and reached us through a Codex-verified handoff of paths, digests and values. We did not access or re-run these files, and none of them is in the frozen packet. The frozen v1 results of Section 6 remain the historical baseline and are not altered by anything here. Table 9 summarizes the four remediation goals against that baseline.

Table 9. The four remediation goals: frozen v1 baseline versus post-freeze v2 evidence. Scopes differ by row and are stated in the subsections; no row implies external generalization or deployment readiness.

Goal	Frozen v1 baseline (MEASURED, 2026-09-24)	Post-freeze v2 (POST-FREEZE (HANDOFF), 2026-09-25)
1. Padding and parity	Fixed padding to 32 slots $\times$ 512 tokens; C – B = 204.373 ms at p50 on CPU; shared-v1 parity 80/80 at atol = rtol = 1e-5	Dynamic padding to the longest live candidate (bounded by the manifest budget and 512) in Python and TypeScript; one dynamic graph ran widths 64 and 128 with finite, identical live outputs; Direct 32 $\times$ 64 p50 19.750145 ms (< 20 ms gate, narrowly); sequence-padding delta p50 243.613376 ms frozen shared-v1 re-verified read-only, 80/80, max abs diff 0.0
2. Data and calibration	Single-label split (182/182 answerable rows call_tool); temperature at the 0.05 search bound; calibrated test ECE 0.2233	Balanced pilot-v2: 2,000 rows, four answerable classes (462/462/461/461), 154 unanswerable; vector scaling v3 with interior per-class temperatures (0.669–0.810); fitted abstention threshold 0.0; calibrated Direct test: accuracy 1.0, macro-F1 1.0 (constant-majority 0.101732), coverage 0.92, selective risk 0.0, ECE 0.036557
3. Diffusion head	Pooled conditioning, 4 DDIM steps; answerable accuracy 0.011–0.060 on the single-label split	Token-level cross-attention over encoder states, cosine schedule, 64 training / 8 inference steps; strictly decreasing epoch loss (1.276957 $\rightarrow$ 0.439098); synthetic test accuracy and macro-F1 1.0 (random-choice 0.25), coverage 0.92, selective risk 0.0, Brier 0.0136773, NLL 0.106455
4. Mind2Web input	Teacher-forced history and full candidate text; 31,769 of 37,512 attempts (84.6902%) over 512 tokens; 0 accepted	Label-free structured DOM compression (≤ 15 tokens per candidate, last 2 history actions, exact-token cap ≤ 512); 37,512 attempts, 0 overflow, 0 abstentions, 37,512 accepted; 323 correct among 5,553 recalled rows, conditional selection accuracy 0.0581668, under an uncalibrated 0.55/ k + 0.5 gate

7.1 Dynamic padding and re-verified parity

The Python package and the TypeScript SDK now pad each request to the longest live tokenized candidate, bounded by the manifest budget and by 512 tokens; the TypeScript timing record’s `sequence_length` field holds the actual feed width. A real ONNX Runtime CPU regression executed one dynamic graph at widths 64 and 128 with finite and identical live outputs. A fresh local CPU shape benchmark (`reports/onnx-shape-decomposition-v2.json`; 80 samples per cell; zero failures) gives Direct 32 slots $\times$ 64 tokens inference + readback p50 19.750145 ms, Direct 4 live slots $\times$ 64 tokens p50 2.672417 ms, Direct 32 slots $\times$ 512 tokens p50 263.363521 ms, and a sequence-padding delta of 243.613376 ms at p50. The design gate of under 20 ms for the 32 $\times$ 64 cell passes narrowly. These are local CPU ONNX Runtime reference values on one machine, not a deployment guarantee, and they are not directly comparable with the frozen v1 cells (Table 6), which used 240 samples and different session settings. The frozen shared-v1 bundle was re-verified read-only, not modified: `reports/shared-v1/parity-v2-readonly.json` records 20 cases, 80 samples, maximum absolute score and answerability differences of 0.0, pass. The scope remains graph-partition parity only. One boundary must be stated: the handoff digest of `parity-v2-readonly.json` is byte-identical to the SHA-256 of the frozen packet member `reports/shared-v1/parity.json`, which we recomputed from the packet. That file carries no timestamp, so a deterministic re-run of the same cases would be byte-identical—but so would a copy; the handoff alone does not distinguish the two, and the re-verification claim rests on the implementer’s statement.

7.2 Balanced synthetic data and vector-scaling calibration

The pilot-v2 synthetic set has 2,000 rows, of which 154 are unanswerable; the answerable labels are balanced across four classes (`call_tool` 462, `clarify` 462, `refuse` 461, `respond_directly` 461). Its 200-row test split has 184 answerable rows (46, 47, 46, 45) and 16 unanswerable rows, so a constant-majority predictor reaches macro-F1 0.101732 on this distribution. Calibration v3 (`vons.calibration/v3`) is vector scaling with per-class temperatures and biases, bounds $[0.1, 10]$ and an L2 penalty of 0.01 on log-temperature. The fitted temperatures are `call_tool` 0.734359, `clarify` 0.741111, `refuse` 0.810072 and `respond_directly` 0.669444, all strictly interior to the bounds, in contrast to the v1 fit that stopped at its lower bound (Section 6.2). Dynamic abstention is fitted from the calibrated multiclass Brier score with target Brier 2.294946×10^{-6} and answerability threshold 0.5 over 185 covered fit rows of 200; the resulting confidence threshold is 0.0. That threshold is a fitted outcome of the Brier-based procedure on this data, not a hard-coded confidence gate, and it should not be read as “no abstention gate”.

The calibrated Direct test report gives answerable accuracy 1.0, macro-F1 1.0, coverage 0.92, selective risk 0.0, Brier 2.760749×10^{-6} , NLL 0.00091409 and ECE 0.036557. Unlike the frozen split, this split can distinguish learning from a constant predictor (macro-F1 1.0 versus 0.101732). The data are still deterministic, rule-following synthetic rows; the result is not external generalization.

7.3 Token-conditioned Diffusion redesign

The redesigned head conditions on token-level cross-attention over the full encoder sequence states rather than the pooled vector, uses a cosine noise schedule with 64 training diffusion steps and 8 fixed inference steps, and is trained with cross-entropy against continuous label-smoothed soft targets. Twelve epoch mean losses are strictly decreasing: 1.276957, 0.974283, 0.612476, 0.469388, 0.451371, 0.447468, 0.444845, 0.443083, 0.441710, 0.440707, 0.439755, 0.439098. The exported ONNX graph passed the checker with dynamic option and sequence axes and executed finitely in ONNX Runtime at shapes $[1, 4, 64]$, $[1, 4, 128]$, $[1, 8, 64]$ and $[1, 2, 512]$. On the balanced synthetic test the head reaches answerable accuracy and macro-F1 1.0 (random-choice baseline 0.25), coverage 0.92, selective risk 0.0, Brier 0.0136773 and NLL 0.106455; decision-head-only latency is 2.559438 ms at p50, a quantity that excludes the encoder and is neither request nor browser latency. This is synthetic-only evidence: it shows that the redesign learns the synthetic rule where the v1 head did not, and it does not rank the Diffusion head above Direct.

7.4 Mind2Web context compression

The historical v1 result stands as the baseline: 37,512 attempts, 31,769 strict 512-token overflow errors (84.6902%), 5,743 confidence abstentions, 0 accepted (Section 6.3). The post-freeze v2 exact- k matrix is complete across the three official held-out splits and $k \in \{5, 10, 20, 32\}$. A label-free structured DOM compression removes presentation attributes; keeps semantic tag, text, type, aria-label and id; caps each candidate at 15 tokens; keeps only the last two history actions; and applies an exact-token hard cap of 512. Across all 37,512 v2 attempts: 37,512 accepted, 0 errors, 0 abstentions, 0 overflows (0.0%). Observed prompts ranged from 52 to 512 tokens (maximum candidate 15 tokens, maximum history 2 actions); the largest original, uncompressed prompt was 185,879 tokens. The replay evaluated 5,553 recalled rows and found 323 correct selections, a weighted micro conditional selection accuracy of 0.0581668 (greater than the frozen 0.0); example endpoints are `test_task` at $k = 5$, 0.21875, and `test_domain` at $k = 32$, 0.0272021.

Two scope statements are essential. First, acceptance in v2 uses a candidate-normalized confidence gate of $0.55/k$ together with the 0.5 answerability threshold and is explicitly uncalibrated; zero abstentions is therefore an execution-enablement result—every attempt now reaches inference and yields a decision—not a calibrated-risk claim. Second, the metric is released-candidate action-target selection conditional

on recall, not end-to-end browser task success, and the retriever’s recall ceiling from Section 6.3 is unchanged.

7.5 Implementer checks (handoff-stated)

The handoff records, as implementer checks that we did not re-run: the full Python suite with 291 tests passed plus 50 subtests; Ruff clean on all changed Python files; 16 TypeScript tests passed; the benchmark bundle build passed; and a git diff check passed. No commit or push had been made at the time of the handoff.

8 Discussion

The input budget breaks first. The frozen Mind2Web result points at the input contract before the model. With teacher-forced history and full candidate text, 84.7% of attempts exceeded 512 tokens before inference. On CPU the sequence budget is also the dominant cost: going from 64 to 512 tokens added 204.373 ms at the median, against 12.289 ms for going from 4 to 32 slots. The post-freeze evidence confirms the diagnosis from both sides: label-free context compression removed every overflow, and dynamic padding moved the 32×64 CPU cell under 20 ms (POST-FREEZE (HANDOFF)). What remains is the retriever’s recall ceiling and an uncalibrated acceptance gate; the resulting conditional selection accuracy of 0.0581668 is a first non-zero measurement, not a capability claim.

The frozen synthetic benchmark cannot measure learning; the balanced one can, within its rules. Because every answerable row of the frozen split shares one label, it cannot separate a learned policy from a constant one, and the v1 Diffusion head’s failure on it is informative only about that configuration. The balanced pilot-v2 split separates the two (constant-majority macro-F1 0.101732 versus 1.0 for both heads, POST-FREEZE (HANDOFF)), but the rows are deterministic rule-following synthetic data, so the ceiling it measures is rule recovery, not decision quality on real workflows.

Calibration: from a boundary fit to an interior fit. The frozen pipeline ran in the intended order on a disjoint split, but its temperature stopped at the search boundary and the calibrated ECE was set by the answerability factor; we regarded calibration as not established. Vector scaling v3 on the balanced split fits interior per-class temperatures and reaches test ECE 0.036557 (POST-FREEZE (HANDOFF)). Its fitted abstention threshold of 0.0 is an outcome of the Brier-targeted procedure on data where every covered prediction is correct; it says nothing about how the gate would behave on rows the model gets wrong, so calibration remains established only on synthetic data.

Direct versus Diffusion is still not adjudicated. On the frozen split quality is uninformative; on the balanced split both heads reach 1.0 (POST-FREEZE (HANDOFF)), so the split does not separate them either. The matched-shape CPU difference is small but includes architecture, the redesigned head has a decision-head-only latency but no request-level or browser cell, and no equal-effort tuning record exists.

What holds, within scope. A shared-encoder bundle with both heads fits under a 64 MiB model-asset design target and partitions exactly on CPU, re-verified read-only after the freeze. The Direct pipeline runs from synthetic data through calibration to evaluation, and with balanced data and vector scaling it produces an interior calibration fit. A historical Direct request configuration produced 500 successful timed samples per requested browser provider on one machine. With compressed context, every Mind2Web attempt reaches inference. The host boundary keeps consequential actions out of the model’s hands by construction, although its live behaviour is unmeasured.

9 Limitations

1. The only quality data for Vons heads are synthetic, and the answerable label is constant.

2. The Mind2Web diagnostic is evaluation-only and was not independently re-run. It uses a lexical retriever with teacher-forced history, and its result is dominated by the token budget.
3. Browser timings come from a historical graph under requested, unverified providers on one Mac, with no independent process launches and no process or GPU memory measurement.
4. CPU shape results cover four cases on one machine.
5. Calibration covers one seed of one head, and its temperature reached the search boundary.
6. The frozen packet carries no Git revision (`git_revision = null` at capture) and no immutable checkpoint identity for the pilot predictions; historical runtime code identity is limited wherever a raw report does not bind it, and no later repository commit can be bound to these measurements retroactively.
7. The public contract is not fully harmonized between Python and TypeScript.
8. No adapter audit log from live workflows exists, so KASI boundary behaviour in deployment is unmeasured.
9. The checks recorded by the implementer at the freeze (241 Python tests passed with 50 subtests; lint exit 0; 10 SDK tests passed; browser, benchmark and parity builds exit 0) and after it (291 tests plus 50 subtests; 16 TypeScript tests; POST-FREEZE (HANDOFF)) were not re-run for this paper, and unit or mocked tests do not establish browser numerical parity.
10. All post-freeze v2 values were supplied through a Codex-verified handoff of paths, digests and values; we did not access the files, recompute any aggregate or re-run any job. They are not in the frozen packet, and their provenance rests on the handoff record (Appendix B).
11. The handoff digest of the post-freeze parity report equals that of the frozen `parity.json`; a deterministic re-run and a copy are indistinguishable from the digest alone (Section 7.1).
12. The v2 CPU shape benchmark uses 80 samples per cell on one local machine and is not directly comparable with the 240-sample v1 cells; the < 20 ms gate passes narrowly and is a local reference, not a deployment guarantee.
13. The balanced synthetic results (both heads at 1.0; ECE 0.036557; fitted threshold 0.0) are deterministic rule-following data and do not measure decision quality on real workflows; the Diffusion redesign has no request-level, browser or Mind2Web measurement.
14. The v2 Mind2Web result uses an explicitly uncalibrated $0.55/k + 0.5$ acceptance gate, so zero abstentions is execution enablement rather than calibrated risk; conditional selection accuracy of 0.0581668 is released-candidate selection under the unchanged lexical-retriever recall ceiling, not browser task success, and no train-split overlap audit exists.

10 Open Gates and Future Work

Table 10 records the implementer’s gate status at the evidence freeze; it was not re-audited for this paper.

Table 10. Release-gate status recorded at the evidence freeze.

Gate	Recorded status
G0 bundle manifest and provenance	implemented and verified
G1 real WASM inference	implemented; parity harness ready; execution open
G2 WebGPU and memory	Direct controlled cells measured; matrix open
G3 shape and padding	CPU alternative implemented and measured; browser matched cells open
G4 Mind2Web	measured under evaluation-only authorization; public and training rights restricted
G5 attribution and training inclusion	preparation complete; inclusion open
G6 vendor workflow comparison	protocol frozen; comparison not run
G7 public release review	release hold
G8 distribution licence	proposal only at freeze; terms selected by the author afterwards (Section 11)
G9 browser extension channel	architecture proposal only

The post-freeze evidence bears on G3 (padding) and on the synthetic and Mind2Web rows of the open work below, but no gate was re-audited or closed for this version; the table records the freeze-time status.

Open measurement work. Required next steps are: Diffusion browser cells; a completed browser numerical-parity matrix; process and GPU memory; timing of the current SDK and shared-v1; score support; DDIM step counts 2 and 8 with additional seeds; lifecycle and offline checks; matched-padding browser cells; complete-case accuracy and end-to-end task success on Mind2Web; a train-split overlap audit; an equal-effort Direct/Diffusion tuning record; Diffusion calibration; immutable checkpoint identity; live KASI audit logs; and the vendor workflow comparison.

Community participation (PLANNED). The author intends to open Vons to outside reading, trial, experience reports, reproduction and contribution. None of this exists yet as a measured activity: there are no external users, reviews, reproductions or hosted demos to report. Any future experience report would be anecdotal and would not count as research validation unless it was collected under a stated protocol with denominators.

11 Availability, Licensing and Disclosure

Article. This article and its original figures are licensed under CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>), which includes commercial reuse under its terms. The licence does not cover the Vons software, quoted source excerpts or third-party assets, raw benchmark data, or private verification records.

Software and models. At the evidence freeze, the Python package metadata declared Apache-2.0 and a proposed alternative licence had not been adopted; the TypeScript package had no licence field, and the shared-v1 bundle carried no licence or notice payload. The release assessment recorded at the freeze is a hold on public release of the software and models (gate G7, Table 10), with every scientific gate open. Separately, and after the freeze (AUTHOR DECISION), the author selected source-only distribution of the Vons software under custom source-available terms: noncommercial use is free, and enterprise and commercial use require a separate agreement. These terms are not an OSI open-source licence and do not cancel grants already made under earlier terms; their licence text and notices are not part of the frozen evidence. Publishing source does not validate model deployment, and none of the results above changes with it.

Code and model locations (AUTHOR DECISION, access not verified). The author has designated <https://github.com/inlevel9-com/Vons>, <https://huggingface.co/INLEVEL9> and <https://huggingface.co/INLEVEL9/Vons> as the distribution locations. Commits, pushes and uploads are performed by

the author personally, and links from the source repository README back to this paper and to the Technical Report remain withheld pending review. Their public-access and upload status could not be verified from the drafting environment; we make no claim about which files are available at any of them.

Publication and listing status (**AUTHOR DECISION status as stated by the author, not re-verified from the typesetting environment**). arXiv submission 8127259 is on hold, and this artifact pass does not itself update or publish it. The Chrome Web Store listing for “Vons — Local Decisions” version 0.1.0 is publicly available and was independently verified on 2026-09-25 at <https://chromewebstore.google.com/detail/vons-%E2%80%94-local-decisions/cbmjkkoojinfgafahncmjlfmicinide>. This establishes listing availability only; it is not a scientific result and says nothing about model quality or production readiness, and it closes no gate of Section 10.

Data. Mind2Web data were used evaluation-only, were excluded from training and are not redistributed; only aggregate reports and hashes are held in the evidence packet. The release status of the synthetic pilot data (v1 and pilot-v2) is undecided. The frozen evidence packet and the post-freeze v2 evidence files (indexed by `reports/goal-v2/evidence-summary.json`) are held by the author; the latter are not part of the packet.

Models used and contributions. The encoder is `google/bert_uncased_L-4_H-256_A-4` [13]. Qwen, Gemma and Muse Glimmer were used only in the historical suitability screen, as comparison records rather than labels. For Muse Glimmer, the upstream model card and LICENSE state Apache-2.0, and a separate upstream usage policy lists prohibited uses and states that the model is not intended for individuals under 18 [28]. The local conversion has no licence field and no mapping to an upstream revision, so its exact upstream revision is not established, and we draw no legal conclusion. Implementation contributions are attributed only to recorded outputs: Claude Code (the `metrics/v2` and `calibration/v2` fixes, with a scoped check of 25 passed tests and 13 subtests); TRAE (a browser raw-sample summary tool accepted with 71 scoped tests, and a read-only licence inventory); a local Qwen model (one bounded static review of the Mind2Web predictor, whose three findings were dispositioned as not reproduced, rejected against an existing guard, and one accepted diagnostic improvement); and Codex (integration, the shared bundle and its parity, Mind2Web acquisition through replay, the browser verifier and orchestration).

Writing assistance. This manuscript was drafted with the assistance of Claude Opus 5.5 and revised for publication with the assistance of Claude Fable 5.1. The assistants verified hashes, recomputed aggregates from saved predictions, checked references and wrote text; they did not generate or alter any measured value. The author is responsible for the content.

12 Conclusion

Vons is a bounded decision contract with a host-only authorization boundary, a compact shared-encoder model with two heads, and an evidence discipline that keeps measured, historical, derived, design-target, planned, unavailable and post-freeze quantities apart. The frozen v1 evidence is mostly negative and is preserved here as the baseline: an uninformative single-label split, a boundary calibration fit, and a Mind2Web diagnostic in which the input budget, not the model, failed first. The post-freeze v2 evidence, reported separately and not independently rerun by us, shows that each of those failures was addressable within the contract: dynamic padding with re-verified parity, an interior calibration fit on balanced synthetic data, a Diffusion redesign that learns the synthetic rule, and context compression that lets every Mind2Web attempt reach inference with a first non-zero conditional selection accuracy. None of this establishes external generalization, calibrated deployment, browser task success or head

superiority; those remain the open gates of Section 10, and this version does not announce readiness for release or production use.

A Superseded Metric Values

An earlier metrics version (v1) scored the same saved predictions differently: it ignored covered unanswerable rows in selective risk and divided the Brier score by the option count. Table 11 lists the superseded values next to the v2 values used in this paper (MEASURED, HISTORICAL).

Table 11. Superseded v1 and current v2 metric values on the same saved predictions.

Report	Selective risk v1 → v2	Brier v1 → v2	ECE v1 → v2
Direct, seed 7	0.0000 → 0.0900	0.0035 → 0.0139	0.3008 → 0.2163
Direct, seed 17	0.0000 → 0.0900	0.0024 → 0.0097	0.1991 → 0.1154
Direct, seed 27	0.0000 → 0.0900	0.0032 → 0.0129	0.2479 → 0.2881
Diffusion, seed 7	0.8214 → 0.8387	0.2330 → 0.9318	0.4327 → 0.4483
Diffusion, seed 17	0.8000 → 0.8103	0.2343 → 0.9370	0.4537 → 0.4602
Diffusion, seed 27	0.9091 → 0.9091	0.2342 → 0.9368	0.5088 → 0.5088

B Provenance and Traceability

Table 12 lists the identifiers that pin the evidence used in this paper. Table 13 maps each section, table and figure to the evidence-packet files it rests on. Row-level bindings (file, JSON pointer or source line, SHA-256 and value) are kept in the author’s audit ledger, which is available for review on request; the *Key* column is the ledger key.

Table 12. Provenance identifiers. Frozen-evidence rows were verified before drafting; post-freeze rows reproduce the digests supplied in the handoff and were not verified by us.

Item	Identifier
Evidence archive publication-evidence-v2.zip, SHA-256	f2fb06c2275a60e6ec6c5b89938123da599b9979bb10d7e65a5667f6ca69152c4
Packet manifest packet.json, SHA-256	7cc5b19cd0e7691919d632d431ff1101d5a5b26a7c11c104c9fa83c132cd5318
Packet schema and freeze time	vons.publication-evidence/v2; 2026-09-24T08:41:25.776377+00:00
Encoder and revision	google/bert_uncased_L-4_H-256_A-4 at 387825ce42dbb39b87911cdf8e383ee3b25184f8
Synthetic test split, SHA-256	e5fdd950a35593b97d9799435a8739177c1668d2ad7d30687546add0bc4de1bc
Synthetic calibration split, SHA-256	a986ab870f9d54d3905db1f3c71ea4e004922a5bdf43fe822e4c503d7dd3689
Mind2Web dataset and dataset revision	osunlp/Mind2Web at 17ece8eb89862368edc0cc806acee6fca5163474
Mind2Web repository revision	33bd95caeee7bba22dd08ecc935845e15c5e5dc7
Mind2Web test archive, SHA-256	8f5fbe72afab942fe97cdf7fb397e179885d89b5c16862288e9a14bc6d41ca89
CPU runtime and platform	ONNX Runtime 1.30.0; macOS-26.6.2-arm64-arm-64bit
<i>Post-freeze v2 (POST-FREEZE (HANDOFF))</i>	<i>digests as supplied in the handoff, 2026-09-25; not in the packet</i>
v2 evidence index evidence-summary.json	39f53771bbf4ffe454dbac09412baf0f371804e6442bb197eaf9879afbfc640a
v2 CPU shape benchmark onnx-shape-decomposition-v2.json	10d1f962d7809b87d4aef64a00ae5804c6cb2aaf1ad037120e1eede1989c1c0a
Shared-v1 read-only parity parity-v2-readonly.json	d680d0f7f1b02f7dc10b39b40b6ff4c6341fe7cc1c317cb4f0a256cbe63b6d5d
v2 calibrated Direct test report	156c0ce009271f93f09f6e778e28248e97fd7c8edd1389886d04ab75990189d7
Diffusion redesign export manifest	bf8a7480c43c0895901a2f30c83316a3cbcd6a0023e3d2a8f363623a5b5aab
Diffusion redesign ONNX graph	7de207c298766ee8eca6921e5f659e33b89689cb6dce38df0f7342bb6339d5c4
Diffusion redesign synthetic test report	b9db4e59e7300c87dea55b17976f3fbc6b15e9dd25bda9d69f35bb6b490517c1
Mind2Web v2 prediction index	55f471d1998ecbf14b8de88a9f22d69b5e47e362dd0c09517f9c91c65ebb04e9
Mind2Web v2 replay summary	95c4464ca5095149c6a6d4805b799fcaa511fef9d3d77f89d82e715c63059fb6

Table 13. Traceability map from manuscript locations to evidence-packet sources.

Key	Location	Evidence sources
M01	Abstract: synthetic pilot	reports/quality-summary-v2.json; data/generated/pilot-test.jsonl
M02	Abstract: calibration	reports/pilot-direct-test-calibrated-v2.json; data/generated/pilot-calibration.manifest.json
M03	Abstract: model assets and parity	artifacts/shared-v1/manifest.json; artifacts/provenance/browser-direct/bundle-manifest-v1.json; artifacts/provenance/browser-diffusion/bundle-manifest-v1.json; reports/shared-v1/parity.json
M04	Abstract, §6.6: browser scope	reports/browser-direct-wasm-trae-summary.json; reports/browser-direct-webgpu-trae-summary.json
M05	Abstract: Mind2Web totals	reports/mind2web/evaluation-replay-v1/summary.json
M06	Abstract: CPU shape decomposition	reports/onnx-shape-decomposition.json
M10	§3: general decision contract	vons/contract.py; sdk/typescript/src/onnx.ts; packet.json
M11	§3, Table 1: KASI host boundary	docs/KASI_INTEGRATION.md; docs/JEV_COMPARISON_PROTOCOL.md; vons/kasi.py; docs/PUBLICATION_REVIEW.md; vons/contract.py; packet.json
M12	§4: encoder, heads, shared bundle	reports/pilot-direct-test-calibrated-v2.json; docs/ORCHESTRATION_STATUS.md; packet.json; artifacts/shared-v1/manifest.json
M13	§2: vendor workflow evaluation status	evals.typesafe.ai; typesafe.ai blog; reports/quality-summary-v2.json; docs/JEV_COMPARISON_PROTOCOL.md; packet.json
M20	§5.1: verification and non-reruns	packet.json; reports/quality-summary-v2.json
M21	§5.2: synthetic data composition	reports/quality-summary-v2.json; data/generated/pilot-calibration.manifest.json; data/generated/pilot-test.jsonl; data/generated/pilot-calibration.jsonl
M22	§5.3: metric definitions	reports/quality-summary-v2.json; vons/evaluation.py; vons/training.py; docs/PUBLICATION_REVIEW.md
M23	§5.4: Mind2Web protocol	reports/mind2web/data-provenance-v1.json; reports/mind2web/evaluation-replay-v1/summary.json; reports/mind2web/prediction-run-v1.json; tools/predict_mind2web.py; packet.json
M24	§5.5: timing protocols	reports/onnx-shape-decomposition.json; reports/browser-direct-wasm-trae-summary.json; reports/browser-direct-webgpu-trae-summary.json; docs/ORCHESTRATION_STATUS.md
M25	§5.6: local-model screen protocol	reports/local-model-alignment-v1.json

Key	Location	Evidence sources
T1, T2	Table 2: synthetic metrics	reports/quality-summary-v2.json
M30	§6.1: synthetic interpretation	reports/quality-summary-v2.json; reports/pilot-direct-test-calibrated-v2.json; data/generated/pilot-test.jsonl
T3	Table 3: calibration	reports/pilot-direct-test-calibrated-v2.json
M31, M47	§6.2: calibration interpretation and order	reports/pilot-direct-test-calibrated-v2.json; packet.json; docs/ORCHESTRATION_STATUS.md; vons/calibration.py
T4	Figure 1, Table 4: Mind2Web cells	reports/mind2web/evaluation-replay-v1/summary.json
M35	§6.3: Mind2Web totals and consequences	reports/mind2web/evaluation-replay-v1/summary.json; packet.json
T5	Table 5: shared-v1 bundle	artifacts/shared-v1/manifest.json
M32	§6.4: bundle and parity	artifacts/shared-v1/manifest.json; artifacts/provenance/browser-direct/bundle-manifest-v1.json; artifacts/provenance/browser-diffusion/bundle-manifest-v1.json; reports/shared-v1/parity.json
T6	Figure 2, Table 6: CPU shape cells	reports/onnx-shape-decomposition.json
M34	§6.5: shape differences	reports/onnx-shape-decomposition.json; packet.json
T7	Table 7: historical browser	reports/browser-direct-wasm-trae-summary.json; reports/browser-direct-webgpu-trae-summary.json
M33, M46	§6.6, §4: browser limits and runtime version	packet.json; internal review record (not public); docs/ORCHESTRATION_STATUS.md; browser summaries
T8	Table 8: local-model screen	reports/local-model-alignment-v1.json; artifacts/ollama-models.json
M36	§6.7: alignment audit	reports/local-model-alignment-v1.json
M40	§8: discussion	reports/mind2web/evaluation-replay-v1/summary.json; reports/onnx-shape-decomposition.json; docs/ORCHESTRATION_STATUS.md; packet.json
M41	§5.1, §9: provenance and recorded checks	packet.json; reports/orchestration-checkpoint-v2.json
M42	§11: models used and contributions	internal review record (not public); docs/ORCHESTRATION_STATUS.md; docs/MODEL_ATTRIBUTION.md; Muse Glimmer LICENSE and USAGE_POLICY (upstream); artifacts/ollama-models.json
M43	§10, Table 10: gates and open work	docs/COMPLETION_PLAN.md; packet.json
M44	Title page, §11: licensing and data	author decision after freeze; docs/PUBLICATION_REVIEW.md; pyproject.toml; packet.json; sdk/typescript/package.json;

Key	Location	Evidence sources
M45	Table 12: provenance identifiers	artifacts/shared-v1/manifest.json; reports/mind2web/data-provenance-v1.json packet.json; reports/quality-summary-v2.json; data/generated/pilot-calibration.manifest.json; reports/pilot-direct-test-calibrated-v2.json; reports/mind2web/data-provenance-v1.json; reports/onnx-shape-decomposition.json
P01	Abstract, Table 9, §7.1: dynamic padding, v2 CPU cells, read-only parity	reports/goal-v2/evidence-summary.json; reports/onnx-shape-decomposition-v2.json; reports/shared-v1/parity-v2-readonly.json (handoff digests)
P02	Abstract, Table 9, §7.2: pilot-v2 composition, calibration v3, calibrated Direct test	reports/goal-v2/evidence-summary.json; calibrated Direct test report (handoff digest 156c0ce0...)
P03	Abstract, Table 9, §7.3: Diffusion redesign	reports/goal-v2/evidence-summary.json; export manifest (bf8a7480...); ONNX graph (7de207c2...); synthetic test report (b9db4e59...)
P04	Abstract, Table 9, §7.4: Mind2Web v2 matrix and replay	reports/goal-v2/evidence-summary.json; prediction index (55f471d1...); replay summary (95c4464c...)
P05	§7.5, §9: implementer checks after the freeze	handoff statement (not a packet file)
TA1	Table 11: superseded metric values	reports/pilot-direct-test.json; reports/pilot-direct-seed17-test.json; reports/pilot-direct-seed27-test.json; reports/pilot-diffusion-test.json; reports/pilot-diffusion-seed17-test.json; reports/pilot-diffusion-seed27-test.json; reports/quality-summary-v2.json

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